Project: Worked examples
Title: Tolerance stack-up (1D)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

A linear tolerance loop: worst-case limits, root-sum-square and modified RSS limits, predicted spread and capability against a stated requirement, contribution Pareto and a Monte Carlo prediction.

WHAT IT CALCULATES	RELATION
Worst case, every contributor at its limit	$T_{wc} = \Sigma\lvert\text{half-bands}\rvert$
Root sum of squares, the statistical stack	$T_{rss} = \sqrt{\Sigma\ \text{half}^2}$
Modified RSS, a factor on the RSS (Bender-style)	$T_{mrss} = \text{factor} \times \text{RSS}$
Predicted standard deviation of the stack, from each contributor's distribution	$\sigma = \sqrt{\Sigma\ \sigma_i^2}$
Process capability, taken from the nearer limit	$C_{pk} = \min(\text{USL} - \mu, \mu - \text{LSL})/(3\sigma)$
Monte Carlo check, every contributor sampled from a fixed seed and summed with its sign	$\text{stack} = \Sigma\ s_i\ x_i$

Basis: Technical product documentation and specification[1]; Dimensioning and tolerancing[2]; General tolerances (where a contributor's band is a general tolerance)[3].

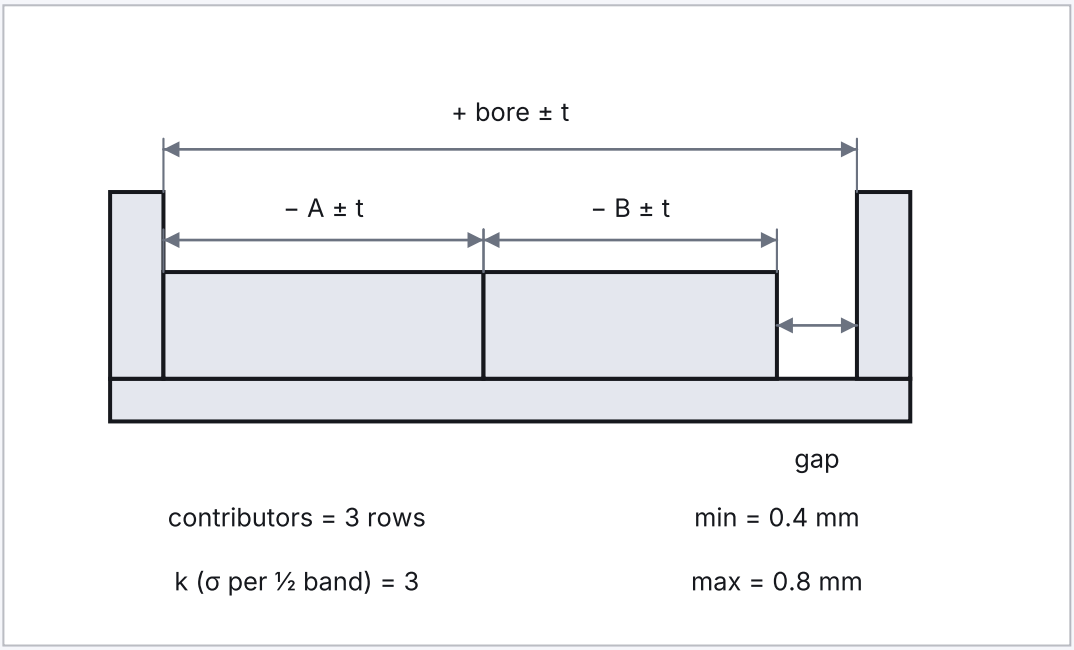


Figure 1 — A linear stack (representative; the contributors as listed): the bore adds, the parts subtract, the gap closes the loop. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
contributors	Contributors in the loop	3 row(s)		drawing: DRG-HSG-001 rev B	any
minimum	Minimum acceptable result	0.4	mm	spec: SPEC-BRG-7 §4.2	-1000000 … 1000000 mm
maximum	Maximum acceptable result	0.8	mm	spec: SPEC-BRG-7 §4.2	-1000000 … 1000000 mm
k	Standard deviations per half-band (normal contributors)	3		assumption: Each part at ±3σ within its band.	1 … 6
f_{rss}	Modified-RSS factor (Bender-style)	1.5		assumption: Bender factor 1.5.	1 … 2
cpk_{target}	Required C_{pk}	1.33		assumption: Usual capability target.	1 … 2
n_{mc}	Monte Carlo samples	10000		assumption: Default sample count.	1000 … 100000
seed	Monte Carlo seed	1		assumption: Fixed seed for reproducibility.	0 … 4295000000
notation	Drawing notation	BS 8888		assumption: Metric drawing.	any

TABLE ROW	VALUES
contributors[1]	name = Housing bore depth; direction = adds; nominal = 40 mm; upper = 0.1 mm; lower = 0 mm
contributors[2]	name = Bearing width; direction = subtracts; nominal = 20 mm; upper = 0 mm; lower = -0.12 mm
contributors[3]	name = Spacer; direction = subtracts; nominal = 19.5 mm; upper = 0.02 mm; lower = -0.02 mm

3. Assumptions

A1. Every contributor acts along the one axis of the stack. (A linear stack: no angular or datum-shift effects beyond what the user enters as a contributor.)
A2. Statistical figures assume contributors vary independently about their mid-limit (plus any stated mean shift), each with the distribution stated for it. (The worst case is a guarantee; every statistical figure is a prediction about a population.)
A3. A position or profile contributor contributes ± half its tolerance zone, plus half the stated MMC bonus. (Diametral zones are bilateral about the true position.)
A4. k (Standard deviations per half-band (normal contributors)) is taken as 3. (Each part at ±3σ within its band.)
A5. f_{rss} (Modified-RSS factor (Bender-style)) is taken as 1.5. (Bender factor 1.5.)
A6. cpk_{target} (Required C_{pk}) is taken as 1.33. (Usual capability target.)
A7. n_{mc} (Monte Carlo samples) is taken as 10000. (Default sample count.)
A8. seed (Monte Carlo seed) is taken as 1. (Fixed seed for reproducibility.)
A9. notation (Drawing notation) is taken as BS 8888. (Metric drawing.)

4. Working

[7] c1 = + 40 +0.1/0

= mid 40.05 mm, ± 0.05 mm, σ 0.0166667 mm (normal)

= 40.05 mm

Housing bore depth

[8] c2 = - 20 +0/-0.12

= mid 19.94 mm, ± 0.06 mm, σ 0.02 mm (normal)

= -19.94 mm

Bearing width

[9] c3 = - 19.5 ±0.02

= mid 19.5 mm, ± 0.02 mm, σ 0.00666667 mm (normal)

= -19.5 mm

Spacer

[10] x_{nom} = Σ (sign × nominal)

= 40 - 20 - 19.5

= 0.5 mm

Nominal result

[11] x_{mean} = Σ (sign × mid-limit)

= 40.05 - 19.94 - 19.5

= 0.61 mm

Mean result

[12] t_{wc} = Σ half-bands

= 0.05 + 0.06 + 0.02

= 0.13 mm

Worst-case tolerance (±)

[13] $x_{wc,min}$ = x_{mean} - t_{wc}

= 0.61 mm − 0.13 mm

= 0.48 mm

Worst-case minimum

[14] $x_{wc,max}$ = x_{mean} + t_{wc}

= 0.61 mm + 0.13 mm

= 0.74 mm

Worst-case maximum

[15] t_{rss} = √Σ half²

= √(0.05² + 0.06² + 0.02²)

= 0.080623 mm

Root-sum-square tolerance (±)

[16] $x_{rss,min}$ = x_{mean} - t_{rss}

= 0.61 mm − 0.080623 mm

= 0.52938 mm

RSS minimum

[17] $x_{rss,max}$ = x_{mean} + t_{rss}

= 0.61 mm + 0.080623 mm

= 0.69062 mm

RSS maximum

[18] t_{mrss} = f_{rss} * t_{rss}

= 1.5 × 0.080623 mm

= 0.12093 mm

Modified RSS tolerance (±)

[19] σ = √Σ σ_i^2

= √(0.0166667² + 0.02² + 0.00666667²)

= 0.026874 mm

Predicted standard deviation of the result

[20] m_{min} = $x_{wc,min}$ - minimum
= 0.48 - 0.4
= 0.08 mm
Worst-case margin to the minimum
[21] m_{max} = maximum - $x_{wc,max}$
= 0.8 - 0.74
= 0.06 mm
Worst-case margin to the maximum
[22] cpk = min(x_{mean} - LSL, USL - x_{mean}) / (3 σ)
= min(0.21, 0.19) / (3 × 0.0268742)
= 2.3567
Process capability index C_{pk}
[23] cp = (USL - LSL) / (6 σ)
= (0.8 - 0.4) / (6 × 0.0268742)
= 2.4807
Process capability C_p
[24] dpmo = 10^6 × P(outside)
= 10^6 × Φ-tails at σ = 0.0268742
= 7.7755e-7
Predicted defects per million
[25] yield = 1 - dpmo / 10^6
= 1 - 7.7755e-7 / 10^6
= 100 %
Predicted yield
[26] u_{cpk} = cpk_{target} / cpk
= 1.33 / 2.3567
= 0.56436
Capability utilisation
[27] pareto = largest contributor (statistical)
= Housing bore depth 38.5 %; Bearing width 55.4 %; Spacer 6.15 %
= "Bearing width"
Contribution Pareto
[30] mc_{mean} = mean of 10000 samples (seed 1)
= Monte Carlo
= 0.60987 mm
Monte Carlo mean
[31] mc_{sigma} = sample standard deviation
= Monte Carlo
= 0.027228 mm
Monte Carlo σ
[32] mc_{p05} = 5th percentile
= Monte Carlo
= 0.56589 mm
Monte Carlo 5 %
[33] mc_{p95} = 95th percentile
= Monte Carlo
= 0.65462 mm
Monte Carlo 95 %
[34] mc_{yield} = samples inside the requirement
= 10000 of 10000
= 100 %
Monte Carlo yield

5. Results

SYMBOL	VALUE	DESCRIPTION
x_{nom}	0.5000 mm	Nominal result
x_{mean}	0.6100 mm	Mean result
t_{wc}	0.1300 mm	Worst-case tolerance (±)
$x_{wc,min}$	0.4800 mm	Worst-case minimum
$x_{wc,max}$	0.7400 mm	Worst-case maximum
t_{rss}	0.08062 mm	RSS tolerance (±)
t_{mrss}	0.1209 mm	Modified RSS tolerance (±)
σ	0.02687 mm	Predicted σ
m_{min}	0.08000 mm	Worst-case margin to the minimum
m_{max}	0.06000 mm	Worst-case margin to the maximum
cp	2.4807	C_p
cpk	2.3567	C_{pk}
dpmo	7.7755e-7	DPMO (predicted)
yield	100 %	Yield (predicted)
mc_{mean}	0.6099 mm	Monte Carlo mean
mc_{sigma}	0.02723 mm	Monte Carlo σ
mc_{p05}	0.5659 mm	Monte Carlo 5th percentile
mc_{p95}	0.6546 mm	Monte Carlo 95th percentile
mc_{yield}	100 %	Monte Carlo yield

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	C_{pk} against target	u_{cpk} = 0.56	u_{cpk} ≤ 1	0.56	
PASS	Worst case within requirement	0.48 … 0.74 mm	0.4 … 0.8 mm	—	

7. Applicability

WARNING contributors: fewer than four contributors; RSS and modified RSS are optimistic for so few. [few-contributors]

8. Verification

```
Template mech.tolerance-stack@2.2.1 Content hash 6ec171c7c89dab1e Engine calc-engine/1.0.0
Known-answer cases passed 7/7 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. British Standards Institution. Technical Product Documentation and Specification. BS 8888. BSI.
2. American Society of Mechanical Engineers. Dimensioning and Tolerancing. ASME Y14.5. ASME.
3. International Organization for Standardization. General Tolerances—Part 1: Tolerances for Linear and Angular Dimensions Without Individual Tolerance Indications. ISO 2768-1. ISO.